

Relational Algebra Assignment

- a) Consider the relational database of Figure below, where the primary keys are underlined. Give an expression in the relational algebra to express each of the following queries:

employee (person-name, street, city)

works (person-name, bank-name, salary)

bank (bank-name, city)

manages (person-name, manager-name)

- i. Find the names of all employees in this database who live in the same city as the company for which they work.
- ii. Give all employees of First Bank Corporation a 10 percent salary raise.
- iii. Modify the database so that Harish now lives in Biratnagar.
- iv. Delete all tuples in works relation for employee of First Bank Corporation.

- b) Consider the relational database of Figure below, Where the primary keys are underlined. Give an expression in the relational algebra to express each of the following queries:

Employee (person-name, street, city)

Works (person-name, bank-name, salary)

Bank (bank-name, city)

Manages (person – name, manager– name)

- i. Find the total salary sum of all the banks.
- ii. Modify the database so that Ram now lives in Kathmandu.
- iii. Fine the name. street address, and cities of residence of all employees who work for Nepal World Bank Corporation and earn more than \$10,000 per annum.
- iv. Delete all tuples in works relation for employee of Small Bank Corporation

- c) Consider the following relations for an order-processing database application in a company.

CUSTOMER (Cust#, Cname, City)

ORDER(Order#, Odate, Cust#, ord_Amt)

ORDER_ITEM(Order#, Item#, Qty)

ITEM(Item#, Unit_Price)

SHIPMENT(Order#, Wavehouse#, Ship_date)

WAREHOUSE(Warchouse#, City)

Answer the following queries in relational Algebra

- i. List the order# and ship date for all orders shipped from wavehouse number “W2”
- ii. List the warehouse information for which the customer named ‘Jose Copez’ was supplied his orders
- iii. List the orders that werenot shipped within 30 days of ordering

- iv. List the order # for orders that were shipped from all warehouses that the company has in network.

d) Consider the following database :

Student(sid,name,age)

Has(sid,cid)

College(cid,cname)

Write relational algebra expression to perform the following:

- i. Find average age of student
- ii. Display name of student who studies in “QWERT” college.
- iii. Insert a new student.
- iv. Delete record of “ASDFG” college from college relation.
- v. Display name of students whose name begin from ‘S’.

e) Consider the schema given below and write relational algebraic statements for the followings.

Depositor = {customer _ name, account _ number}

Borrower = { customer_ name, loan _ number}

- i. Find all tuples of loan for amount < 5000.
- ii. Find all customers who have both loan and account.
- iii. Add a field ‘address’ to the relation Depositor.
- iv. Create a view to show only the customer name from the borrower relation.
- v. Delete the field loan_ number from the relation Borrower.

f) By using following schemas write relational algebraic expression and SQL statements.
(Underlined attributes represent Primary key attributes)

EMPLOYEE(EMPNO, NAME, ADDRESS)

PROJECT(PNO,PNAME)

WORKON(EMPNO, PNO)

PART(PARTNO, PARTNAME, QTY_ON_HAND)

USE(EMPNO,PNO,PARTNO,NUMBER)

- i. Listing all the employee details who are not working yet.
- ii. Listing Part Name and Quantity on hand those were used in DBMS project.
- iii. List the Name of the projects that are used by employee from Kathmandu.

g) Consider the following relational database.

job (job-id, job-title, minimum-salary, maximum-salary)

employee (emp-no, emp-name, emp-salary, dept-no)

deposit (acc-no, cust-name, branch-name, amount, account-date)

borrow (loan-no, cust-name, branch-name, amount)

department (dept-no, dept-name)

Write the relational algebra expression for the following queries.

1. Give name of depositors whose branch name starts from 'D'.
2. Give employee name(s) whose salary is between Rs. 20000 to 30000 and department name is Finance.
3. Delete borrower details whose amount is less than 25000.
4. Insert new employee details {'E105', 'Roshan Singh', '60000', 'D109'}.